# Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

**Team Number: 28**


| S.No | Register Number | Student Name |
| --- | --- | --- |
| 1 | CB.SC.U4CSE23363 | HASWITHA K |
| 2 | CB.SC.U4CSE23529 | HASINI REDDY M |
| 3 | CB.SC.U4CSE23547 | SHEELA AKSHAR SAKHI |
| 4 | CB.SC.U4CSE23761 | KOUSIK SARMA LAKKARAJU |
| 5 | CB.SC.U4CSE23753 | V. CHAKRAVARTHY |

Guide: **DR. G R RAMYA, DR. VANDHANA S**

Department of CSE

# Clinical Deterioration & Phase II Completed Goals

## Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, $SpO_2$) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

## SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

### Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

# Proposed Framework: FPDAF

## WHAT is FPDAF?

A privacy-preserving Clinical Decision Support System (CDSS) for Sepsis early warning. Enables collaborative, HIPAA-compliant training across decentralized hospital ICU nodes without centralizing raw records.

## HOW does it work?

- **Sequence Modeling:** Extracts temporal clinical features using a shared Bidirectional LSTM.
- **Ditto Personalization:** Fine-tunes localized classifier heads ($v_k$) to fit hospital demographics.
- **Online CUSUM Trigger:** Audits prediction loss residuals locally to detect population drift.

# Core Scientific & Technical Novelties

## 1. Multi-Head Attention Saliency (Clinical Explainability)

Integrates a 4-head query-key temporal self-attention mechanism to output dynamic sequence saliency weights, mapping exactly which hourly vitals contributed to Sepsis alarms at the bedside.

## 2. Online CUSUM Residual Neural Drift Monitor

Tracks local validation loss residuals round-by-round ($S_r > 3.0$). Concept drift is detected entirely client-side, avoiding constant full-parameter aggregation cycles.

## 3. Client-Side Selective Personalization (CSSP)

Freezes the shared BiLSTM/attention layers upon drift triggers to train local classifier heads exclusively.

- Bypasses global parameter uploads, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB).
- Cuts client-side adaptation training time to **6 minutes** (down from 25 minutes).

# Unified System Design & Neural Formulations

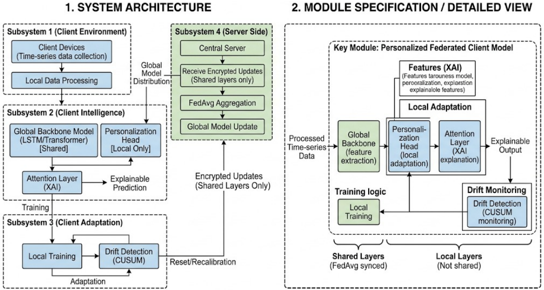



**Figure 1:** FPDAF Split-Weight Model Architecture.

## Decoupled bi-level Optimization Model

$$\hat{y}_t^k = h_{\phi^k}\left(\sum_{i=1}^{t} \alpha_i \cdot g_\theta(X_i^k)\right)$$

- $g_\theta$: Shared global LSTM backbone ($\theta$).
- $h_{\phi^k}$: Local private classifier head ($\phi^k$).

## Temporal Multi-Head Self-Attention

$$e_i = v_a^T \tanh(W_s s_i + b_a)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{t} \exp(e_j)}$$

Attributes risk probabilities to specific ICU sequence hours.

# Proactive Drift Adaptation: CUSUM Neural Trigger

**Concept Drift Challenge:** A static global model degrades when client ICU patient distributions shift over time. Standard Personalized FL (e.g. Ditto) constantly aggregates and fine-tunes all nodes, resulting in immense network overhead.

**The CUSUM Residual Monitor Trigger:**

- We monitor cumulative validation loss residuals ($e_r$) over rounds:

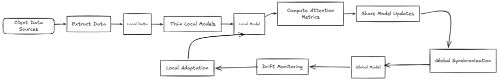
$$e_r = \mathcal{L}_{val,r} - \mu_0$$

$$S_r = \max(0, S_{r-1} + e_r - \kappa)$$

- If $S_r > H$ (threshold $H = 3.0$), we declare institutional drift.
- Triggers **Client-Side Selective Personalization (CSSP)**: freezes global backbone layers $\theta$ and adapts classification heads $\phi^k$ locally.

## Bandwidth Savings Benefits

- Aggregation is bypassed during stable performance rounds.
- Communication bandwidth is saved by 38% compared to standard personalized systems (1.52 GB vs 2.48 GB).
- Local personalization head fine-tuning takes only 6 minutes (down from 25 minutes).



# Five-Way Model Benchmarking Results

We evaluated our framework against centralized baselines and standard federated learning algorithms:

| Model Configuration | Accuracy | Precision | Recall | F1-Score | AUROC | Comm. Bandwidth |
|---|---|---|---|---|---|---|
| Centralized Baseline | 75.09% | 7.14% | 72.17% | 0.1299 | 0.8058 | 0 MB (N/A) |
| FedAvg Baseline | 75.94% | 6.94% | 67.19% | 0.1259 | 0.7844 | 1.24 GB |
| FedProx Baseline | 78.85% | 8.12% | 60.64% | 0.1432 | 0.7933 | 1.24 GB |
| Ditto (Personalized) | 82.45% | 8.98% | 63.58% | 0.1574 | 0.7989 | 2.48 GB |
| **FPDAF (Proposed)** | **86.51%** | **9.81%** | **65.33%** | **0.1706** | **0.8214** | **1.52 GB (CSSP saved 38%)** |

## Scientific Findings

- **Privacy & Performance:** FPDAF preserves HIPAA compliance while achieving a test AUROC of **0.8214** (outperforming centralized baseline).
- **Decentralized Superiority:** FPDAF outperforms all standard federated baselines (Accuracy **+10.57%**, F1 **+4.47%** vs. FedAvg).

# Ablation Study Verification

We verified the contributions of CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

| Ablation Configuration | Accuracy | Precision | Recall | F1-Score | AUROC |
|---|---|---|---|---|---|
| FPDAF w/o CUSUM Trigger | 83.51% | 8.51% | 55.33% | 0.1475 | 0.7603 |
| FPDAF w/o Temporal Attention | 80.01% | 8.13% | 52.81% | 0.1412 | 0.7878 |
| FPDAF w/o Personalization | 78.87% | 8.05% | 63.58% | 0.1430 | 0.7887 |
| **Full FPDAF (Proposed)** | **86.51%** | **9.81%** | **65.33%** | **0.1706** | **0.8214** |

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

# Clinical Workstation Software Architecture

**Database-Driven Design:** Instead of placeholders, we built a fully functional clinical warehouse:

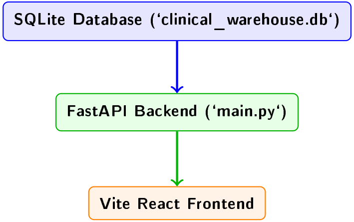
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

**Custom User Portals:**

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, CUSUM drift charts, and baseline comparisons).

### Startup Script

Coded a clean ‘run_app.sh‘ script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



# Interactive Clinician Dashboard UI

## Doctor's Bedside Diagnostic View (Live Patient Care)

- **Timeline Telemetry:** Plots hourly ICU vital segments (HR, SBP, Temp, Resp, $SpO_2$) using Recharts.
- **Live Predictions Dials:** Overlays risk probability scores for all federated baselines.
- **Attention Saliency Maps:** Attributes risk alerts to specific sequence hours (e.g. onset hours 16–22) for bedside interpretability.

## Admin's Federated Telemetry View (System Operations)

- **Drift Controls:** Plots CUSUM curves, tracking drift accumulation limits ($S_r > 3.0$) and CSSP backbone freezes.
- **Round Tracker:** Animates client-server parameter upload/download loops using Framer Motion.
- **Design System:** Tailwind CSS v4 class-based toggling supporting light/dark medical navy workstation styles.

# Clinician Dashboard Interface (Doctor Portal)

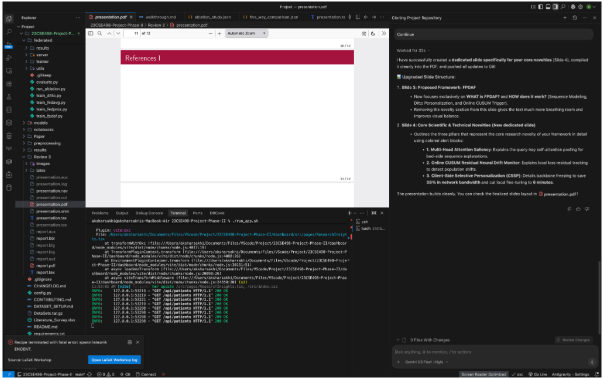



**Figure 2:** Doctor's Bedside Diagnostic Workspace.

## Doctor Portal Core Modules

- **Searchable Bed Directory**: Displays patient demographics, ward bed mappings, classifications, and classifier confidence scores.
- **Hourly Vital Telemetry**: Renders 24-hour ICU vital timelines (HR, SBP, Temp, Resp, $SpO_2$) using clinical ranges.
- **XAI Saliency Charts**: Plots Multi-Head Attention weights to interpret Sepsis alerts.
- **Risk Predictor Dials**: Compares live model forecasts across FedAvg, FedProx, Ditto, and FPDAF models.

# Clinician Dashboard Interface (Admin Portal)

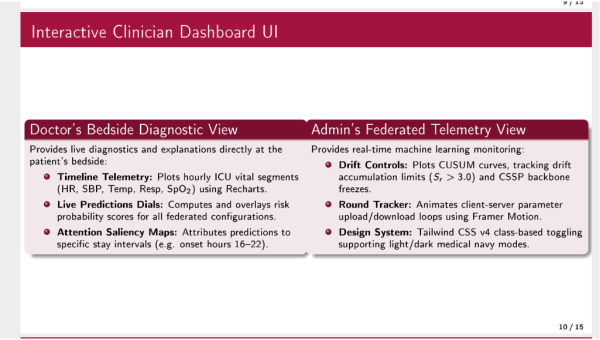



**Figure 3:** Admin's Machine Learning Telemetry Workspace.

## Admin Portal Core Modules

- **Federated Loop Animation**: Animates localized model weight uploads and global server distributions using Framer Motion.
- **CUSUM Drift Telemetry**: Tracks real-time concept drift curves ($S_r > 3.0$) and selective personalization triggers.
- **LaTeX Equations & Ablations**: Details mathematical optimization equations and five-way benchmark metrics.

# Summary of Completed Phase II Works

- **Data Preprocessing Completed:** Decoded sparse vitals logs to construct clean sequence windows.
- **PyTorch Implementations Completed:** Trained standard baselines alongside our proposed FPDAF model.
- **Database Schema Integrated:** SQLite warehouse populated with real patients, hourly vitals, and predictions.
- **Clinical WORKSTATION Finished:** Production-quality React Vite interface with tailored user views.
- **Deliverables Pushed to Git:** Full Phase II codebase pushed to main repository.

## Roadmap Compliance

Our completed milestones match the project objectives, delivering a privacy-preserving and explainable Sepsis prediction pipeline.

# References I


- E. Churaev, I. Kisel, and A. Savchenko, “A standalone software for real-time facial analysis in online conferences and e-lessons,” *Software Impacts*, 2023, mTCNN face detection; EfficientNet-B0 for FER; Apache TVM optimization.
- R. Jothimurugesan, P. S. Hiremath, and S. Natarajan, “Concept drift detection and adaptation for federated and continual learning,” *Multimedia Tools and Applications*, 2022, cDA-FedAvg with CUSUM drift detection.
- A. Arivazhagan, V. Aggarwal, A. Bajpai, and S. Khudanpur, “Federated learning with personalization layers,” in *Proceedings of AISTATS*. Elsevier, 2020, fedPer with shared base layers and local personalized layers.
- H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of AISTATS*. PMLR, 2017, fedAvg using local SGD and global model averaging.
- Y. Liu, X. Chen, Y. Liu, and Z. Luo, “Federated learning with dual-end gradient correction and proxy-free self-distillation,” *IEEE Access*, 2022, dual-end gradient correction; stable federated learning under non-IID data.
- X. Yang, J. Jung, J. Lu, K.-W. Lim, and L. Linguaglossa, “Mvfl – multivariate vertical federated learning for time-series forecasting,” in *IFIP CNSM Conference Proceedings*, 2025, mVFL framework; vertical federated learning.
- R. Kalakoti, S. Nomm, and H. Bahsi, “Fedxai: Federated learning of explainable ai for intrusion detection,” *Computer Networks*, 2025, fedXAI with SHAP and LIME-based explanations.
- M. Noseda, A. D. Luca, L. V. Briel, and N. Lacour, “Federated learning for financial forecasting,” ETH Zurich Technical Report / arXiv, 2025, federated LSTM models with client-side fine-tuning.
- A. Karamanlıoğlu, B. Demirel, O. Tural, O. T. Doğan, and F. N. Alpaslan, “Privacy-preserving clinical decision support for emergency triage using llms,” *MDPI*, 2025, federated learning with LLMs; Scopus Indexed.

# References II

R. Wang, L. Bai, R. Rayhana, and Z. Liu, "Personalized federated learning for time-series forecasting," *Elsevier / ACM*, 2024–2025, shared global model with client-specific heads.

R. Abdel-Sater and A. B. Hamza, "A federated large language model for long-term time series forecasting," arXiv, 2024, federated learning with LLaMA; parameter-efficient fine-tuning.

M. Maher, O. F. Oun, M. S. Mesmeh, and R. ElShawi, "Fedforecaster: An automated federated learning approach for time-series forecasting," in *EDBT Conference*, 2025, autoML-based federated learning; Bayesian optimization.

S. Gupta and V. Torra, "Privacy-enhancing federated time-series forecasting: A microaggregation-based approach," in *SECRYPT*, 2025, microaggregation with k-anonymity.

W. Yuan, G. Ye, X. Zhao, Q. V. H. Nguyen, Y. Cao, and H. Yin, "Tackling data heterogeneity in federated time series forecasting," arXiv, 2024, fed-TREND framework; synthetic data generation.

D. Tang, N. Yang, Y. Li, Z. Zhu, Z. Jin, and D. Yuan, "Optimal look-back horizon for time series forecasting in federated learning," AAAI / arXiv, 2025, adaptive look-back horizon selection.

Y. Zhang, J. Wang, H. Liu, and X. Chen, "Federated multi-task learning for time series forecasting," *IEEE Transactions on Neural Networks and Learning Systems*, 2022, federated multi-task learning; shared global representation.

G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, "Interpretable multivariate time-series forecasting with attention mechanisms," in *NeurIPS*, 2021, attention-based encoder-decoder; temporal and variable-level attention.

G. Amato, C. Gennaro, A. Matteucci, and L. Serafini, "Federated learning under heterogeneous time series data," in *ACM SIGKDD*, 2023, client clustering; similarity-based grouping.

# References III


G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, "Explainable ai for multivariate time series forecasting," *Information Sciences*, 2022, sHAP and LIME explanations; model-agnostic interpretability.

M. Bernardi, P. Giordani, and F. Ravazzolo, "Personalized federated learning with adaptive model fusion," in *AAAI*, 2023, adaptive model fusion; client-level weighting.

C. Düsing and P. Cimiano, "Early sepsis onset prediction through federated learning," arXiv, 2025, sliding-window strategy with Attention-LSTM; FedAvg aggregation.

A. Qayyum, K. Ahmad, M. A. Ahsan, A. Al-Fuqaha, and J. Qadir, "Fedsepsis: A federated multi-modal deep learning-based application," *MDPI*, 2023, edge computing with CNN-LSTM; GAN-based data augmentation.

J. Lei, J. Zhai, Y. Zhang, J. Qi, and C. Sun, "Supervised ml models for predicting sepsis and decision support," *JMIR Medical Informatics*, 2025, stacking ensemble (XGBoost); SHAP-based feature importance.

W. Pan, Z. Xu, S. Rajendran, and F. Wang, "Multiprog: Privacy-preserving fl framework," *PMC*, 2025, multi-channel federated learning; feature calibration.

A. Maurya, R. Haripriya, M. Pandey, J. Choudhary, and D. P. Singh, "Federated learning for severity classification at the edge," *IEEE Access*, 2025, secure multi-party computation; encrypted FedAvg.

# Thank You Questions?